

COMBATTING NEIGHBORHOOD THEFTS

- A DATA ENGINEERING APPROACH





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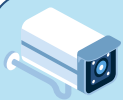
Overview
Introduction – Problem Statement and Importance
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OVERVIEW

Neighborhood thefts pose a significant challenge to the safety and well-being of our community. In this project, we leveraged the power of data engineering via GCP to uncover insights and develop strategies to address this pressing issue





INTRODUCTION

The Problem:

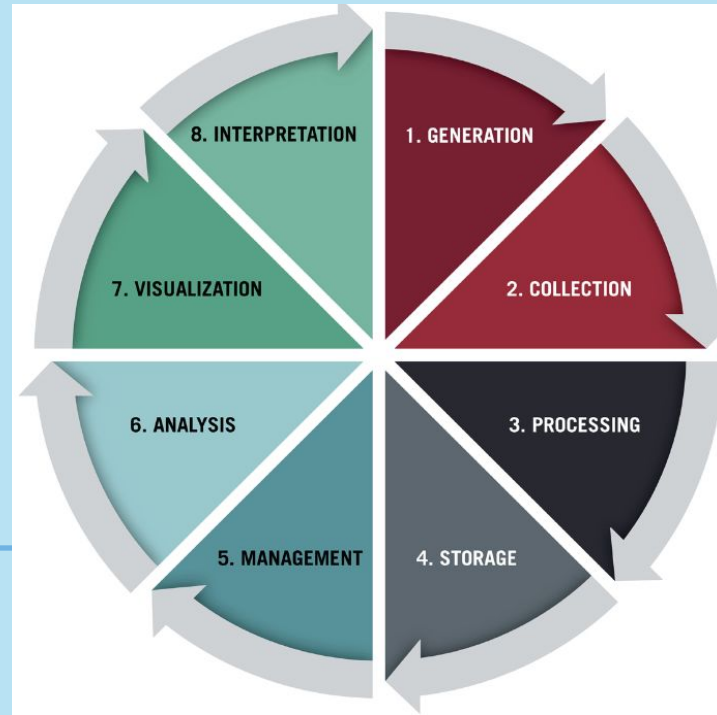
With rising temperatures during the summer, so do the property thefts in the neighborhood. With longer days and vacations, homes become vulnerable, and criminals seize the opportunity.

Importance:

By understanding the underlying patterns and factors contributing to neighborhood thefts, we can empower local authorities and community leaders to implement targeted solutions that enhance safety and improve the quality of life for Denton residents.



IMPLEMENTATION THROUGH DATA LIFECYCLE



Identified
Sources in
the form of
static
datasets

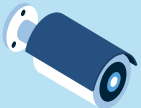
Open Refine

Google Cloud Storage

Google Cloud
Platform

Big Query, Hive & Spark SQL





DATA GENERATION & COLLECTION

We identified several key sources for generating data related to neighborhood thefts. Some of them include:

- Police Department Records
- Community Reports
- Sensor data

By utilizing these diverse data sources and collection methods, we were able to assemble 3 static datasets that provided valuable insights into the patterns and contributing factors behind neighborhood thefts in Denton.

- <https://communitycrimemap.com/>
- https://data.syr.gov/datasets/5b53f40abbc54c559ca991c96816cc17_0/about
- https://www.dallasopendata.com/Public-Safety/Police-Incidents/qv6i-ri7/about_data



Data generation and collection. (n.d.).
Shutter Stock.





DATA PROCESSING USING OPEN REFINE

The first screenshot shows the OpenRefine interface with the 'police incidents' dataset. The 'Filter' panel on the left lists various actions, including 'Rename column Month1 of Occurrence to Month of Occurrence' and 'Remove column Drug Related Istevenincident'. The second screenshot shows the 'Filter' panel with a list of actions such as 'Remove 51 rows', 'Remove 97 rows', and 'Mass edit 16,998 cells in column Weapon Used'. The third screenshot shows the 'Filter' panel with a list of actions including 'Create project', 'Remove column Service Number ID', and 'Remove column Incident Address'.

We employed **OpenRefine** to ensure the quality and integrity of our data, we Data Cleaning and Preparation Techniques we used:

OpenRefine helped us remove duplicates, standardize data formats, and handle missing values, by ensuring a clean and consistent dataset for further analysis.

We also utilized OpenRefine's advanced features to detect and resolve inconsistencies in the data, enhancing the overall data quality.




OpenRefine police incidents [Permalink](#)

Open... Export Help

Facet / Filter
Undo / Redo 15 / 15
19529 rows

Extensions Wikibase

Refresh
Reset all Remove all
Show as: rows records Show: 5 10 25 50 100 500 1000 rows
« first « previous 1 next » last »

Blank records per column
change
10 choices Sort by: name count
Call (911) Problem 61
City 32
Family Offense 1
Gang Related Offense 17079
Modus Operandi (MO) 51
State 107
UCR Disposition 25
Victim Type 18
Weapon Used 17261
Zip Code 10
Facet by choice counts

All	Incident Number w/year	Year of Incident	Call (911) Problem	Type of Incident	Type Location	Type of Property	
☆	1. 090983-2022	2023	DAEV-DIST ARMED ENCOUNTER VEH	BURGLARY OF HABITATION -NO FORCED ENTRY	Single Family Residence - Occupied	Residential Property Occupied/Vacant	421
☆	2. 046070-2022	2023	40/01 - OTHER	ASSAULT (AGG) -DISCH FIREARM OCC BLDG/HOUSE/VEH (AGG)	Highway, Street, Alley ETC	Motor Vehicle	419
☆	3. 270941-2017	2023	09V - UUMV	UNAUTHORIZED USE OF MOTOR VEH - AUTOMOBILE	Highway, Street, Alley ETC	Outdoor Area Public/Private	105
☆	4. 097663-2022	2023	09V-01 UUMV JUST OCRD	THEFT OF PROP > OR EQUAL \$2,500 <\$30K (NOT SHOPLIFT) PC31.03(e4A)	Single Family Residence - Occupied	Other	445
☆	5. 015577-2017	2023	09V - UUMV	UNAUTHORIZED USE OF MOTOR VEH - AUTOMOBILE	Parking Lot (All Others)	Parking Lot	601
☆	6. 146239-2022	2023	**PD REQUESTED BY FIRE	NATURAL DEATH (NO OFFENSE)	Single Family Residence - Occupied	Residential Property Occupied/Vacant	421
☆	7. 012324-2022	2023	20 - ROBBERY	ROBBERY OF INDIVIDUAL (AGG)	Bar/NightClub/DanceHall ETC.	Other	105
☆	8. 228924-2021	2023	PSE/09 - THEFT	THEFT OF PROP (AUTO ACC) <\$100 - (NOT EMP)	Parking Lot (All Others)	Motor Vehicle	206
☆	9. 004880-2023	2023	11R/01 - BURG OF RES	BURGLARY OF HABITATION - FORCED ENTRY	Apartment Residence	Apartment Complex/Building	452
☆	10. 251329-2018	2023	09V - UUMV	UNAUTHORIZED USE OF MOTOR VEH - TRUCK OR BUS	Outdoor Area Public/Private	Outdoor Area Public/Private	436
☆	11. 020225-2019	2023	58 - ROUTINE INVESTIGATION	UNAUTHORIZED USE OF MOTOR VEH - AUTOMOBILE	Parking (Business)	Motor Vehicle	304
☆	12. 030967-2023	2023	PSE/09 - THEFT	THEFT OF PROP > OR EQUAL \$100 <\$750 (NOT SHOPLIFT)	Apartment Complex/Building	Residential Property Occupied/Vacant	102

Screenshot – OpenRefine interface Selected blank facets as part of data cleaning process.





DATA STORAGE

Upon cleaning and processing the data, we utilized Google Cloud Storage buckets to store the datasets.

Cloud Storage provided us with a centralized and easily accessible repository for the processed data, facilitating seamless data sharing and integration across various GCP services throughout our data engineering pipeline.



Google Cloud PROJECT-GROUP9-NTHEFTS buckets

Cloud Storage Bucket details datasets_nt9

Location: us-south1 (Dallas) Storage class: Standard Public access: Not public Protection: Soft Delete

OBJECTS CONFIGURATION PERMISSIONS PROTECTION LIFECYCLE OBSERVABILITY INVENTORY REPORTS OPERATIONS

Folder browser datasets_nt9

Upload files Upload folder Create folder Transfer data Manage holds Edit retention Download Delete

Filter by name prefix only Filter objects and folders Show Live objects only

	Name	Size	Type	Created	Storage class	Last modified	Public access	
☐	Communitycrimemap.csv	69.5 KB	text/csv	Apr 22, 2024, 3:04:46 PM	Standard	Apr 22, 2024, 3:04:46 PM	Not public	Download More actions
☐	Policeincidents.csv	4.2 MB	text/csv	Apr 22, 2024, 3:04:56 PM	Standard	Apr 22, 2024, 3:04:56 PM	Not public	Download More actions
☐	SPDPCComplaints.csv	7.6 KB	text/csv	Apr 22, 2024, 3:05:09 PM	Standard	Apr 22, 2024, 3:05:09 PM	Not public	Download More actions

3 files successfully uploaded

Processed data was stored in Google Storage Buckets



Screenshot - Data Storage Bucket management in Google Cloud Storage

Screenshot - Datasets uploaded into Google Cloud Storage Buckets.

DATA MANAGEMENT

As part of a crucial step in our data engineering pipeline, we leveraged Google Cloud's Dataproc service to set up a robust Hadoop ecosystem for managing and processing the neighborhood theft data.

Dataproc allowed us to set-up a Hadoop cluster tailored to our project's requirements.

We configured the cluster with the following specifications:

- Cluster Name: "dataproc-Hadoop-spark-cluster"
- Region and Zone: us-central1
- Manager Node: e2-standard-4 machine type with 128 GB disk size
- Worker Nodes: 2 nodes with 128 GB disk size
- By utilizing Hadoop environment (HDFS) managed by Dataproc, we were able to focus on the data engineering tasks.



DATA MANAGEMENT

The Dataproc cluster served as the foundation for various stages of our data engineering workflow:

- ❖ **Data Processing:** We used the Hadoop Distributed File System (HDFS) within the Dataproc cluster to store the refined datasets, ensuring scalable and reliable data storage.
- ❖ **Data Analysis:** Leveraging Apache Spark and Apache Hive, both of which are pre-installed on the Dataproc cluster, we were able to perform advanced analytics and extract valuable insights from the neighborhood theft data.
- ❖ **Data Management:** The Dataproc cluster provided us with a centralized and easily accessible platform for managing the data and running queries to support the project's objectives.

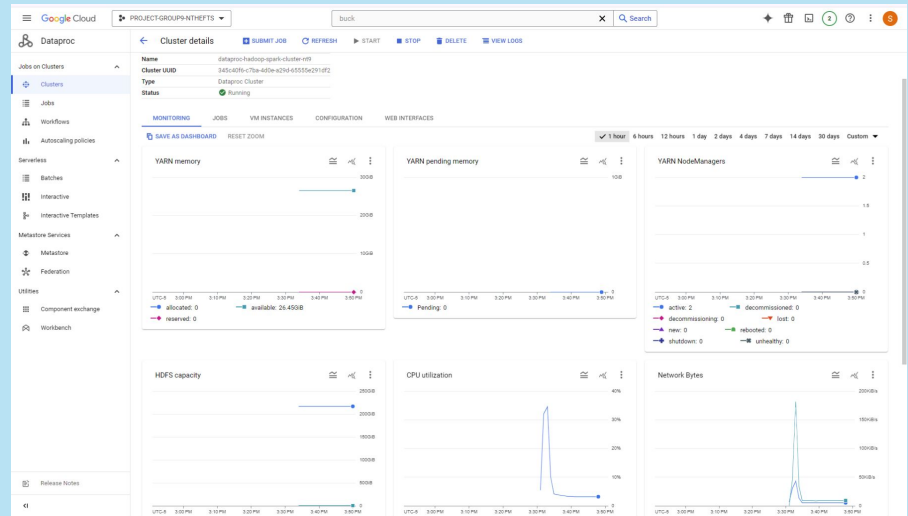
By deploying the Dataproc cluster within the Google Cloud ecosystem, we were able to integrate it with other GCP services, such as Cloud Storage and BigQuery.



Integration with GCP:

The screenshot shows the Google Cloud console interface for the 'PROJECT-GROUP-NTHFTS' project. The 'Compute Engine' section is active, displaying the 'VM instances' page. The left sidebar lists various services like Virtual machines, Instance templates, Machine images, and more. The main content area shows a table of VM instances with columns for status, name, zone, recommendations, internal IP, external IP, and connect options. Below the table, there are several 'Related actions' cards such as 'Explore Backup and DR', 'View billing report', 'Monitor VMs', and 'Explore VM logs'.

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
Running	dataproc-hadoop-spark-cluster-rtf-m	us-central1-a			10.128.0.4 (red)	34.133.202.143 (red)	SSH
Running	dataproc-hadoop-spark-cluster-rtf-a-0	us-central1-a			10.128.0.3 (red)	34.122.221.160 (red)	SSH
Running	dataproc-hadoop-spark-cluster-rtf-a-1	us-central1-a			10.128.0.2 (red)	34.173.194.196 (red)	SSH



DATA MANAGEMENT WITH BIGQUERY

To manage and query the processed data effectively, we utilized BigQuery -GCP's fully-managed enterprise data warehouse solution.

By creating an external table in BigQuery, we were able to directly access the data stored in our Cloud Storage bucket, streamlining the data management process and enabling us to run sophisticated SQL queries.

The screenshot displays the Google Cloud BigQuery console. On the left, the 'Explorer' sidebar shows a project named 'PROJECT-GROUP9-NTHEFTS' with various resources like queries, notebooks, and datasets. The main panel shows a SQL query titled 'Community Crime Query Insights'. The query is a multi-part SQL statement that counts incidents by class and location, groups them by crime type, and calculates cumulative incidents over time. Below the query, the 'Query results' section is visible, showing a table with 12 rows and 3 columns: 'Class', 'crime_count', and 'Location_Name'. The results are sorted by crime count in descending order.

Row	Class	crime_count	Location_Name
1	Burglary from Motor Vehicle	83	Parking/drop Lot/garage
2	Theft	62	Residence/home
3	Theft	59	Grocery/supermarket
4	Shoplifting	51	Department/discount Store
5	Theft	40	Department/discount Store
6	Burglary from Motor Vehicle	27	Residence/home
7	Theft	22	Parking/drop Lot/garage
8	Burglary - Residential	19	Residence/home
9	Shoplifting	16	Grocery/supermarket
10	Theft	13	Specialty Store
11	Shoplifting	10	Specialty Store
12	Theft	9	Conv Store

Screenshot of querying using Bigquery and results

Screenshot by Suprathika Vangari

Querying on 3 datasets using BigQuery.

The screenshot displays the Google Cloud BigQuery web interface. The top navigation bar includes the Google Cloud logo, a project selector set to 'PROJECT-GROUP9-NTHEFTS', a search bar, and various utility icons. The left sidebar shows the Explorer view with a search bar and a tree of resources including 'project-group9-nthefts', 'community_crime_map', and 'bigquery-public-data'. The main panel is titled 'Crime Count with respect to Class and Location' and contains a SQL query. Below the query, the 'Query results' section is active, showing a table with 17 rows of crime data. The table has columns for Row, Class, crime_count, and Location_Name. The bottom of the interface shows a 'Job history' section.

```
1 SELECT Class, COUNT(*) AS crime_count, Location_Name
2 FROM `project-group9-nthefts.community_crime_map`
3 GROUP BY Class, Location_Name
4 ORDER BY crime_count DESC
5 LIMIT 30;
6
```

Row	Class	crime_count	Location_Name
1	Burglary from Motor Vehicle	83	Parking/drop Lot/garage
2	Theft	62	Residence/home
3	Theft	59	Grocery/supermarket
4	Shoplifting	51	Department/discount Store
5	Theft	40	Department/discount Store
6	Burglary from Motor Vehicle	27	Residence/home
7	Theft	22	Parking/drop Lot/garage
8	Burglary - Residential	19	Residence/home
9	Shoplifting	16	Grocery/supermarket
10	Theft	13	Specialty Store
11	Shoplifting	10	Specialty Store
12	Theft	9	Conv Store
13	Burglary from Motor Vehicle	9	Highway/road/alley/street/sid...
14	Theft	8	Other/unknown
15	Theft	7	Const Site
16	Burglary - Commercial	7	Residence/home
17	Theft	5	Highway/road/alley/street/sid...

Screenshot – Resultant Crime Class according to crime count and location

By Suprathika Vangari



ADVANCED ANALYSIS WITH HIVE & SPARK



To perform advanced analysis on the neighborhood theft data, we utilized Apache Hive, a data warehouse software that provides a SQL-like interface for querying and managing structured data

```
ssh.cloud.google.com/v2/ssh/projects/project-group9-nthefts/zones/us-central1-a/instances/dataproc-hadoop-spark-cluster-nt9-m?authuser=1&hl=en_US&projectNumber=77502636633&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/project-group9-nthefts/zones/us-central1-a/instances/dataproc-hadoop-spark-cluster-nt9-m?authuser=1&hl=en_US&projectNumber=77502636633&useAdminProxy=true
SSH-in-browser
suprathikavangari@dataproc-hadoop-spark-cluster-nt9-m:~$ cd mkdir nthefts 9
-bash: cd: too many arguments
suprathikavangari@dataproc-hadoop-spark-cluster-nt9-m:~$ mkdir nthefts9
suprathikavangari@dataproc-hadoop-spark-cluster-nt9-m:~$ mkdir nthefts9
mkdir: cannot create directory 'nthefts9': file exists
suprathikavangari@dataproc-hadoop-spark-cluster-nt9-m:~$ cd nthefts9
suprathikavangari@dataproc-hadoop-spark-cluster-nt9-m:~/nthefts9$ ps -ef | grep hadoop
root      997      1  20:31 ?        00:00:21 /usr/bin/java -XX:+AlwaysPreTouch -Xms800m -XX:+HeapDumpOnOutOfMemoryError -XX:HeapDumpPath=/var/crash/google-dataproc-agent.hprof -Dja
va.util.logging.config.file=/etc/google-dataproc/logging.properties -cp /usr/local/share/google-dataproc/agent.jar /etc/hadoop/conf:/usr/lib/hadoop/lib/*:/usr/lib/hadoop/
hadoop-hdfs://usr/lib/hadoop-hdfs/lib/*:/usr/lib/hadoop-hdfs/*:/usr/lib/hadoop-mapreduce/*:/usr/lib/hadoop-yarn/lib/*:/usr/lib/hadoop-yarn/*:/usr/local/share/google-dataproc/lib/*
-XX:CrashOnOutOfMemoryError.com.google.cloud.hadoop.services.agent.AgentMain /usr/local/share/google-dataproc/startup-script.sh /usr/local/share/google-dataproc/post-hdfs-startup-script.sh
mapred    5707      1  20:31 ?        00:00:42 /usr/lib/jvm/temurin-8-jdk-amd64/bin/java -Dproc_historyserver -Dmapred.jobsummary.logger=INFO,RFA -XX:UseConcMarkSweepGC -XX:PrintGCDateStamps
s -XX:PrintGCDateStamps -XX:PrintGCDetails -Dyarn.log.dir=/var/log/hadoop-mapreduce -Dyarn.log.file=hadoop-mapred-historyserver-dataproc-hadoop-spark-cluster-nt9-m.log -Dyarn.home.dir=/usr/l
ib/hadoop-yarn -Dyarn.root.logger=INFO,console -Djava.library.path=/usr/lib/hadoop/lib/native -Xmx4000m -Dhadoop.log.dir=/var/log/hadoop-mapreduce -Dhadoop.log.file=hadoop-mapred-historyserver
INFO,NullAppender org.apache.hadoop.mapreduce.v2.lib.HistoryServer
dataproc-hadoop-spark-cluster-nt9-m.log -Dhadoop.home.dir=/usr/lib/hadoop -Dhadoop.id.str=mapred -Dhadoop.root.logger=INFO,RFA -Dhadoop.policy.file=hadoop-policy.xml -Dhadoop.security.logger=
INFO,NullAppender org.apache.hadoop.mapreduce.v2.lib.HistoryServer
yarn      5812      1  20:31 ?        00:00:16 /usr/lib/jvm/temurin-8-jdk-amd64/bin/java -Dproc_timelineserver -XX:UseConcMarkSweepGC -XX:PrintGCDateStamps -XX:PrintGCDateStamps -XX:Print
GCDetails -Djava.util.logging.config.file=/etc/hadoop/conf/yarn-timelineserver.logging.properties -Dyarn.log.dir=/var/log/hadoop-yarn -Dyarn.log.file=hadoop-yarn-timelineserver-dataproc-hadoop
-spark-cluster-nt9-m.log -Dyarn.home.dir=/usr/lib/hadoop-yarn -Dyarn.root.logger=INFO,console -Djava.library.path=/usr/lib/hadoop/lib/native -Xmx4000m -Dhadoop.log.dir=/var/log/hadoop-yarn -Dh
adoop.log.file=hadoop-yarn-timelineserver-dataproc-hadoop-spark-cluster-nt9-m.log -Dhadoop.home.dir=/usr/lib/hadoop -Dhadoop.id.str=yarn -Dhadoop.root.logger=INFO,RFA -Dhadoop.policy.file=hado
op-policy.xml -Dhadoop.security.logger=INFO,NullAppender org.apache.hadoop.yarn.server.applicationhistoryservice.ApplicationHistoryServer
yarn      5839      1  20:31 ?        00:00:26 /usr/lib/jvm/temurin-8-jdk-amd64/bin/java -Dproc_resourcemanager -Dservice.libdir=/usr/lib/hadoop-yarn/lib:/usr/lib/hado
op-hdfs://usr/lib/hadoop-hdfs/lib:/usr/lib/hadoop/./usr/lib/hadoop/lib -Xmx4000m -Dyarn.log.dir=/var/log/hadoop-yarn -Dyarn.log.file=hadoop-yarn-resource-manager-dataproc-hadoop-spark-clust
er-nt9-m.log -Dyarn.home.dir=/usr/lib/hadoop-yarn -Dyarn.root.logger=INFO,console -Djava.library.path=/usr/lib/hadoop/lib/native -Dhadoop.log.dir=/var/log/hadoop-yarn -Dhadoop.log.file=hadoop-
yarn-resource-manager-dataproc-hadoop-spark-cluster-nt9-m.log -Dhadoop.home.dir=/usr/lib/hadoop -Dhadoop.id.str=yarn -Dhadoop.root.logger=INFO,RFA -Dhadoop.policy.file=hadoop-policy.xml -Dhadoo
p.security.logger=INFO,NullAppender org.apache.hadoop.yarn.server.resourcemanager.ResourceManager
hive      5844      1  20:31 ?        00:00:26 /usr/lib/jvm/temurin-8-jdk-amd64/bin/java -Dproc_jar -Dhive.log.dir=/var/log/hive -Dhive.log.file=hive-metastore.log -Dhive.log.threshold=INFO -Xm
x4000m -Dproc_metastore -Dlog4j2.formatMsgNoLookups=true -XX:UseConcMarkSweepGC -XX:PrintGCDateStamps -XX:PrintGCDateStamps -XX:PrintGCDetails -XX:ExitOnOutOfMemoryError -Dlog4j.configu
rationFile=hive-log4j2.properties -Djava.util.logging.config.file=/usr/lib/hive/conf/parquet-logging.properties -Dyarn.log.dir=/usr/lib/hadoop/logs -Dyarn.log.file=hadoop-log -Dyarn.home.dir=/
usr/lib/hadoop-yarn -Dyarn.root.logger=INFO,console -Djava.library.path=/usr/lib/hadoop/lib/native -Dhadoop.log.dir=/usr/lib/hadoop/logs -Dhadoop.log.file=hadoop-log -Dhadoop.home.dir=/usr/lib
/hadoop -Dhadoop.id.str=hive -Dhadoop.root.logger=INFO,console -Dhadoop.policy.file=hadoop-policy.xml -Dhadoop.security.logger=INFO,NullAppender org.apache.hadoop.util.RunJar /usr/lib/hive/lib
/hive-metastore-3.1.3.jar org.apache.hadoop.hive.metastore.HiveMetaStore
hdfs      6352      1  20:32 ?        00:00:34 /usr/lib/jvm/temurin-8-jdk-amd64/bin/java -Dproc_namenode -Dhdfs.audit.logger=INFO,NullAppender -Xmx3201m -XX:UseConcMarkSweepGC -XX:PrintGCDat
estamps -XX:PrintGCDateStamps -XX:PrintGCDetails -Dyarn.log.dir=/var/log/hadoop-hdfs -Dyarn.log.file=hadoop-hdfs-namenode-dataproc-hadoop-spark-cluster-nt9-m.log -Dyarn.home.dir=/usr/lib/had
oop-yarn -Dyarn.root.logger=INFO,console -Djava.library.path=/usr/lib/hadoop/lib/native -Dhadoop.log.dir=/var/log/hadoop-hdfs -Dhadoop.log.file=hadoop-hdfs-namenode-dataproc-hadoop-spark-clus
ter-nt9-m.log -Dhadoop.home.dir=/usr/lib/hadoop -Dhadoop.id.str=hdfs -Dhadoop.root.logger=INFO,RFA -Dhadoop.policy.file=hadoop-policy.xml -Dhadoop.security.logger=INFO,NullAppender org.apache
hadoop.hdfs.server.namenode.NameNode
hdfs      6747      1  20:32 ?        00:01:55 /usr/lib/jvm/temurin-8-jdk-amd64/bin/java -Dproc_secondarynamenode -Dhdfs.audit.logger=INFO,NullAppender -Xmx3201m -XX:UseConcMarkSweepGC -XX:
PrintGCDateStamps -XX:PrintGCDateStamps -XX:PrintGCDetails -Dyarn.log.dir=/var/log/hadoop-hdfs -Dyarn.log.file=hadoop-hdfs-secondarynamenode-dataproc-hadoop-spark-cluster-nt9-m.log -Dyarn.ho
me.dir=/usr/lib/hadoop-yarn -Dyarn.root.logger=INFO,console -Djava.library.path=/usr/lib/hadoop/lib/native -Dhadoop.log.dir=/var/log/hadoop-hdfs -Dhadoop.log.file=hadoop-hdfs-secondarynamenode
-dataproc-hadoop-spark-cluster-nt9-m.log -Dhadoop.home.dir=/usr/lib/hadoop -Dhadoop.id.str=hdfs -Dhadoop.root.logger=INFO,RFA -Dhadoop.policy.file=hadoop-policy.xml -Dhadoop.security.logger=IN
FO,NullAppender org.apache.hadoop.hdfs.server.namenode.SecondaryNameNode
hive      7577      1  20:32 ?        00:00:23 /usr/lib/jvm/temurin-8-jdk-amd64/bin/java -Dproc_jar -Dhive.log.dir=/var/log/hive -Dhive.log.file=hive-server2.log -Dhive.log.threshold=INFO -Xm
x4000m -Dproc_hiveserver2 -Dlog4j2.formatMsgNoLookups=true -XX:UseConcMarkSweepGC -XX:PrintGCDateStamps -XX:PrintGCDateStamps -XX:PrintGCDetails -XX:ExitOnOutOfMemoryError -Dlog4j.configu
rationFile=hive-log4j2.properties -Djava.util.logging.config.file=/usr/lib/hive/conf/parquet-logging.properties -Dyarn.log.dir=/usr/lib/hadoop/logs -Dyarn.log.file=hadoop-log -Dyarn.home.dir=/
usr/lib/hadoop-yarn -Dyarn.root.logger=INFO,console -Djava.library.path=/usr/lib/hadoop/lib/native -Dhadoop.log.dir=/usr/lib/hadoop/logs -Dhadoop.log.file=hadoop-log -Dhadoop.home.dir=/usr/lib
/hadoop -Dhadoop.id.str=hive -Dhadoop.root.logger=INFO,console -Dhadoop.policy.file=hadoop-policy.xml -Dhadoop.security.logger=INFO,NullAppender org.apache.hadoop.util.RunJar /usr/lib/hive/lib
/hive-service-3.1.3.jar org.apache.hive.service.server.HiveServer2
spark     7605      1  20:32 ?        00:00:31 /usr/lib/jvm/temurin-8-jdk-amd64/bin/java -cp /usr/lib/spark/conf:/usr/lib/spark/jars/*:/etc/hadoop/conf:/etc/hive/conf:/usr/local/share/goog
le-dataproc/lib/*:/usr/share/java/mysql-connector-java-5.0.8-000.jar org.apache.spark.deploy.history.HistoryServer
suprath+ 11269 10566 0 20:46 pts/0 00:00:00 grep hadoop
suprathikavangari@dataproc-hadoop-spark-cluster-nt9-m:~/nthefts9$
```




```
ssh.cloud.google.com/v2/ssh/projects/project-group9-nthefts/zones/us-central1-a/instances/dataproc-hadoop-spark-cluster-nt9-m?authuser=1&hl=en_US&projectNumber=77502636633&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/project-group9-nthefts/zones/us-central1-a/instances/dataproc-hadoop-spark-cluster-nt9-m?authuser=1&hl=en_US&projectNumber=77502636633&useAdminProxy=true
SSH-in-browser
[+] UPLOAD FILE [+] DOWNLOAD FILE [!] [🔍] [⚙️]

+-----+
1 row selected (23.329 seconds)
0: jdbc:hive2://localhost:10000: SELECT * FROM community_crime_map LIMIT 5;
INFO : Compiling command(queryId=hive_20240423002042_88e5f0f0-89b0-422a-a79b-03748ce38061): SELECT * FROM community_crime_map LIMIT 5
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Semantic Analysis Completed (retrial = false)
INFO : Returning Hive schema: Schema(fieldSchemas:[FieldSchema(name:community_crime_map.class, type:string, comment:null), FieldSchema(name:community_crime_map.incident, type:int, comment:null), FieldSchema(name:community_crime_map.date_time, type:string, comment:null), FieldSchema(name:community_crime_map.location_name, type:string, comment:null), FieldSchema(name:community_crime_map.address, type:string, comment:null), FieldSchema(name:community_crime_map.agency, type:string, comment:null)], properties:null)
INFO : Completed compiling command(queryId=hive_20240423002042_88e5f0f0-89b0-422a-a79b-03748ce38061); Time taken: 0.217 seconds
INFO : Concurrency mode is disabled, not creating a lock manager
INFO : Executing command(queryId=hive_20240423002042_88e5f0f0-89b0-422a-a79b-03748ce38061): SELECT * FROM community_crime_map LIMIT 5
INFO : Query ID = hive_20240423002042_88e5f0f0-89b0-422a-a79b-03748ce38061
INFO : Total jobs = 1
INFO : Launching Job 1 out of 1
INFO : Starting task [Stage-1:MAPRED] in serial mode
INFO : Subscribed to counters: {} for queryId: hive_20240423002042_88e5f0f0-89b0-422a-a79b-03748ce38061
INFO : Session is already open
INFO : Dag name: SELECT * FROM community_crime_map LIMIT 5 (Stage-1)
INFO : Status: Running (Executing on YARN cluster with App id application_1713817921868_0001)

-----+-----+-----+-----+-----+-----+-----+-----+
VERTICES  MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----+-----+-----+-----+-----+-----+-----+-----+
Map 1 ..... container  SUCCEEDED      1          1          0          0          0          0
-----+-----+-----+-----+-----+-----+-----+-----+
VERTICES: 01/01 [----->>>] 100% ELAPSED TIME: 5.23 s
-----+-----+-----+-----+-----+-----+-----+-----+
INFO : Completed executing command(queryId=hive_20240423002042_88e5f0f0-89b0-422a-a79b-03748ce38061); Time taken: 5.915 seconds
INFO : OK
INFO : Concurrency mode is disabled, not creating a lock manager
+-----+-----+-----+-----+-----+-----+-----+-----+
| community_crime_map.class | community_crime_map.incident | community_crime_map.date_time | community_crime_map.location_name | community_crime_map.address |
+-----+-----+-----+-----+-----+-----+-----+-----+
| Class                     | Incident                     | Date/Time                     | Location Name                     | Address                     |
+-----+-----+-----+-----+-----+-----+-----+-----+
| Burglary - Commercial     | 24064480                    | 04/03/2024 03:30 PM           | Residence/home                    | "37xx Prominence Pkwy     |
| Denton                    | 24064479                    | 04/03/2024 05:12 AM           | Residence/home                    | "7xx Hickory St           |
| Burglary - Commercial     | 24063478                    | 04/02/2024 06:55 PM           | Residence/home                    | "1xx Precision Dr         |
| Denton                    | 24060608                    | 03/30/2024 01:00 AM           | Specialty Store                   | "20xx Fort Worth Dr       |
+-----+-----+-----+-----+-----+-----+-----+-----+
5 rows selected (6.187 seconds)
0: jdbc:hive2://localhost:10000>
```

Hive allowed us to define a schema for the data, ensuring a consistent and organized data model for our analysis

Screenshot - Sample HiveQL code snippet for schema definition



In addition to Hive, we leveraged the power of Apache Spark SQL, to develop custom algorithms and queries tailored to our specific analysis needs.



```
ssh.cloud.google.com/v2/ssh/projects/project-group9-nthefts/zones/us-central1-a/instances/dataproc-hadoop-spark-cluster-nt9-m?authuser=1&hl=en_US&projectNumber=77502636633&useAdminProxy=true - Google Chrome
SSH-in-browser
Linux dataproc-hadoop-spark-cluster-nt9-m 5.10.0-0.deb10.16-cloud-amd64 #1 SMP Debian 5.10.127-2-bpo10+1 (2022-07-28) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Apr 22 23:24:06 2024 from 35.235.244.32
suprathikavangari@dataproc-hadoop-spark-cluster-nt9-m:~$ spark-sql
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
ivysettings.xml file not found in HIVE_HOME or HIVE_CONF_DIR, etc/hive/conf.dist/ivysettings.xml will be used
24/04/23 00:45:49 INFO org.apache.spark.SparkEnv: Registering MapOutputTracker
24/04/23 00:45:49 INFO org.apache.spark.SparkEnv: Registering BlockManagerMaster
24/04/23 00:45:50 INFO org.apache.spark.SparkEnv: Registering BlockManagerMasterHeartbeat
24/04/23 00:45:50 INFO org.apache.spark.SparkEnv: Registering OutputCommitCoordinator
Spark master: yarn, Application Id: application_1713817921868_0003
spark-sql> show tables;
default community_crime_map      false
default community_crime_map_1    false
Time taken: 4.026 seconds, Fetched 2 row(s)
spark-sql> SELECT * FROM community_crime_map LIMIT 1;
24/04/23 00:47:12 WARN org.apache.hadoop.hive.ql.session.SessionState: METASTORE_FILTER_HOOK will be ignored, since hive.security.authorization.manager is set to instance of HiveAuthorizerFact
ory.
Class NULL      Crime      Date/Time      Location Name      Address Agency
Time taken: 6.968 seconds, Fetched 1 row(s)
spark-sql> SELECT * FROM community_crime_map LIMIT 5;
Class NULL      Crime      Date/Time      Location Name      Address Agency
Burglary - Commercial  24064480      Theft From Bldg  04/03/2024 03:30 PM      Residence/home  "37xx Prominence Pkwy  Denton
Burglary - Commercial  24064479      Theft From Bldg  04/03/2024 05:12 AM      Residence/home  "7xx Hickory St Denton
Burglary - Commercial  24063478      Theft From Bldg  04/02/2024 06:55 PM      Residence/home  "1xx Precision Dr   Denton
Burglary - Commercial  24060608      Theft From Bldg  03/30/2024 01:00 AM      Specialty Store  "20xx Fort Worth Dr  Denton
Time taken: 0.321 seconds, Fetched 5 row(s)
spark-sql> SELECT * FROM community_crime_map LIMIT 10;
Class NULL      Crime      Date/Time      Location Name      Address Agency
Burglary - Commercial  24064480      Theft From Bldg  04/03/2024 03:30 PM      Residence/home  "37xx Prominence Pkwy  Denton
Burglary - Commercial  24064479      Theft From Bldg  04/03/2024 05:12 AM      Residence/home  "7xx Hickory St Denton
Burglary - Commercial  24063478      Theft From Bldg  04/02/2024 06:55 PM      Residence/home  "1xx Precision Dr   Denton
Burglary - Commercial  24060608      Theft From Bldg  03/30/2024 01:00 AM      Specialty Store  "20xx Fort Worth Dr  Denton
Burglary - Commercial  24062984      Theft From Bldg  03/29/2024 02:00 PM      Residence/home  "13xx Scripture St   Denton
Burglary - Commercial  24059770      Theft From Bldg  03/23/2024 02:05 PM      Residence/home  "34xx Nottingham Dr  Denton
Burglary - Commercial  24048728      Theft From Bldg  03/11/2024 11:00 PM      Residence/home  "30xx Locust St  Denton
Burglary - Commercial  24041704      Theft From Bldg  02/29/2024 02:00 PM      Service/gas Station  "3xx Eagle Dr   Denton
Burglary - Commercial  24039041      Theft From Bldg  02/26/2024 07:00 AM      Residence/home  "5xx Texas St   Denton
Time taken: 0.281 seconds, Fetched 10 row(s)
spark-sql>
```

Snippet of Spark Code and its output



KEY FINDINGS & RECOMMENDATIONS

As data engineers, our role was to establish a robust and efficient data pipeline to support the analysis of neighborhood thefts in Denton.

Through our data engineering efforts, we have laid the foundation for data analysts and scientists to uncover valuable insights and develop targeted solutions.

The data engineering processes we implemented, such as data collection, cleaning, and storage, have provided a high-quality and well-structured dataset for further analysis. Building upon the insights generated from the data pipeline, we have the following recommendations for data analysts and scientists to further explore and implement:

Leverage the community reporting data to establish a neighborhood watch program, empowering residents to actively participate in crime prevention and reporting.



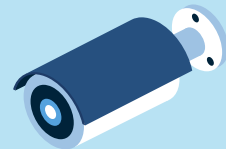
CONCLUSION

As data engineers, our role has been to establish a comprehensive and scalable data pipeline to support the analysis of neighborhood thefts in Denton. By employing GCP and data engineering best practices, we have created a foundation for data analysts and scientists to uncover valuable insights and develop data-driven solutions.

Moving forward, we encourage the data analysts and scientists to fully utilize the data pipeline we have established.

Through the integration of various data sources, including police records, community reports, and the data processing, storage, and management techniques we implemented ensure the reliability, accessibility, and scalability of the data, empowering the data science team to focus on advanced analysis and insightful recommendations.





THANK YOU!

